Type-Aware Failure Prediction for Chart Reasoning in Vision-Language Models

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Abstract

To write (Yash): Abstract: at most 200 words, plain text with no LaTeX commands.

1 Introduction

To write (Yash): Motivation, refined problem statement, contributions, paper outline. Start from paper/mid/main.tex Section 1; drop "and Diagram" (AGENTS.md decision 1).

2 Background and Related Work

To write (Yash): Start from paper/mid/main.tex Section 2 (references.bib is copied from the mid report).

3 Experimental Setup

To write (Shrish): Models and inference, activation caching, probes, splits and bootstrap; the hypothesis can open this section or the introduction.

4 Labelling the Errors

To write (Sanjith): Five-label taxonomy, label counts for both models, and the human agreement check (two annotators kappa 0.421; below the 0.6 gate; adjudicated human labels vs LLM kappa 0.614; results/human_agreement.md).

5 Naturalistic Arm: ChartQA (E1 to E3)

To write (unassigned): E1 binary probe with robustness checks, E2 surface and residualised baselines, E3 misread vs computation on real charts. Start from paper/mid/main.tex Section 5.

6 Synthetic Arm

To write (Yash): Pilots 1 and 2 design and outcomes, missing-vs-shown probes, E3 cross-transfer matrix, grid-line baseline, transfer to ChartQA, and pilot 3 (Table 1).

7 Does the failure type decide the repair? (E4)

We re-query all erroneous value questions in synthetic pilots 1 and 2 (631 for Qwen, 1,015 for LLaVA) to test whether the best repair depends on failure type. Types come from the original answers, independently of probes: misreads of shown bars (S), invented nonzero values for missing bars (F), and missing-bar "0" answers (Z). We score Z as wrong, separately from fabrication.

	Qwen	LLaVA
Answer to the out-of-range year		
Refused	102	0
"0"	76	19
Made-up number	105	281
Unclear	17	0
Shown year read correctly	273	74
Missing-vs-shown probe		
layer	L10	L18
out-of-range vs. shown	1.000 [1.00, 1.00]	0.911 [0.89, 0.93]
made-up vs. correct	1.000	0.922
made-up vs. refused	0.521	–
misread vs. correct	0.494	0.568
Fabrication probe (E3)		
layer	L10	L19
out-of-range vs. shown	1.000 [1.00, 1.00]	0.880 [0.85, 0.90]

Table 1: Pilot 3: each chart is asked about a shown year and, in the same words, about a year outside the shown range. Top: what the models answered to the out-of-range year (300 per model). Bottom: AUROC of the two missing-bar probes, trained on pilots 1 and 2 only and applied unchanged, with 95% figure-bootstrap CIs; layer chosen on pilots 1 and 2.

We compare four repairs: a sampled re-ask of the original question (temperature 0.7); whole-chart upsampling; a question-conditioned crop retaining the target bar and y-axis strip; and a verification prompt requesting chart evidence before the final answer. The latter three use greedy decoding. Upsample and crop share a 1,000,000-pixel budget. Recovery requires a correct read under the original 5% numeric tolerance, or an explicit refusal for a missing bar. The re-ask controls for improvement from another attempt.

Before running the interventions, we preregistered the difference in recovery gains between S and F as the primary contrast on 2 September:

$$\Delta_{\text{crop}} = (r_{S,\text{crop}} - r_{S,0}) - (r_{F,\text{crop}} - r_{F,0}), \quad (1)$$

Original answer	n	Re-ask	Upsample	Crop	Verify
Qwen2.5-VL-7B					
Misread	46	0.11	0.13	**0.26**	0.09
Made-up number	91	0.00	0.00	0.00	**0.14**
"0"	494	0.02	0.02	0.02	**0.69**
crop DiD			+0.261 [+0.140, +0.389]		
LLaVA-NeXT-7B					
Misread	415	0.03	0.01	**0.15**	0.03
Made-up number	368	0.00	0.00	0.00	0.00
"0"	232	0.00	0.00	0.00	0.00
crop DiD			+0.152 [+0.119, +0.189]		

Table 2: E4 recovery rates on initially wrong value answers. Bold marks each row's best repair, excluding zero ties. DiD compares crop's recovery gain on misreads with its gain on made-up numbers. Brackets give 95% CIs from figure bootstraps; missing-bar "0" answers form a separate row.

where r denotes recovery rate. Baseline recovery is zero because we select only errors; the contrast excludes Z. We use 2,000 figure-bootstrap resamples for 95% confidence intervals (CIs).

The crop DiDs are +0.261 for Qwen and +0.152 for LLaVA, with CIs excluding zero as predicted (Table 2). The pre-registered secondary verify DiD is -0.056 [-0.162, +0.053] for Qwen, including zero, and +0.031 [+0.015, +0.050] for LLaVA.

Qwen's best repair is crop for misreads and verify for fabrications and Z answers, recovering 14.3% and 69.2% respectively. Misreads and fabrications select different best repairs in 99% of bootstrap resamples. LLaVA's crop helps misreads, but no repair recovers any of its 600 missing-bar errors. All repairs tie on these errors, making an argmax change uninformative. Re-asking turns 215 of its 232 Z answers into invented numbers.

Crop falls back to upsampling when the bar is missing. The two repairs give identical answers on all 1,185 missing-bar errors across models, so the primary contrast cannot isolate localisation. Comparing crop with upsample isolates this gain: +0.145 [0.111, 0.181] for LLaVA and +0.130 [-0.023, 0.289] for Qwen. Qwen's interval includes zero, with only 46 misreads available.

8 A type-aware controller (E5)

We compare a type-aware controller with binary selective prediction on all 1,200 value questions per model, including correct reads and refusals. Three linear probes on the last prompt-token hidden state score wrong versus right, missing versus shown bar, and misread versus correct read before generation.

Policy	AURC	95% CI
Qwen2.5-VL-7B		
Plain model	0.526	[0.513, 0.538]
Binary abstention	0.179	[0.169, 0.190]
Missing-bar abstention	0.208	[0.191, 0.226]
Controller	0.162	[0.145, 0.180]
Re-look + binary	0.114	[0.103, 0.126]
Oracle	0.056	[0.049, 0.064]
re-look gain (binary)	-0.065	[-0.073, -0.056]
re-look gain (missing-bar)	-0.045	[-0.054, -0.037]
LLaVA-NeXT-7B		
Plain model	0.846	[0.827, 0.864]
Binary abstention	0.645	[0.608, 0.681]
Missing-bar abstention	0.725	[0.686, 0.759]
Controller	0.715	[0.674, 0.750]
Re-look + binary	0.638	[0.600, 0.674]
Oracle	0.497	[0.464, 0.528]
re-look gain (binary)	-0.007	[-0.012, -0.003]
re-look gain (missing-bar)	-0.010	[-0.017, -0.005]

Table 3: E5 AURC on 1,200 value questions per model, with 95% figure-bootstrap CIs. Each re-look gain compares policies with the same abstention ranking: re-look plus binary minus binary, or controller minus missing-bar abstention. A negative difference means re-looking helps. The oracle uses true classes.

Scores use five figure folds: each probe trains on the other four, selecting layer and regularisation within its training folds.

The *controller* re-looks above a misread-score threshold and withholds questions with the highest missing-bar scores first. We choose the repair and threshold to maximise accuracy on four folds, then apply them to the fifth. Baselines include the plain model, binary abstention, missing-bar abstention, and *Re-look + binary*, which uses the controller's re-look rule with binary ranking. The label-aware oracle chooses the best action for each true class and withholds the least recoverable classes first. Re-query outcomes on initially correct answers count harmful repairs against each policy: crop breaks 133 of Qwen's 554 correct reads.

Coverage is the fraction retained; risk is their error rate. AURC averages risk over coverage (lower is better), averaging tied confidence scores. Correct refusals count as correct answers; policy abstention removes questions from coverage. CIs use 1,000 figure-bootstrap resamples.

Adding a re-look lowers AURC under either abstention ranking in both models (Table 3, Figure 1; all four gain CIs exclude zero). The controller beats binary abstention only for Qwen: AURC difference -0.017 [-0.028, -0.004], compared with +0.069 [0.045, 0.094] for LLaVA. Without re-looking, binary ranking also outperforms missing-bar ranking

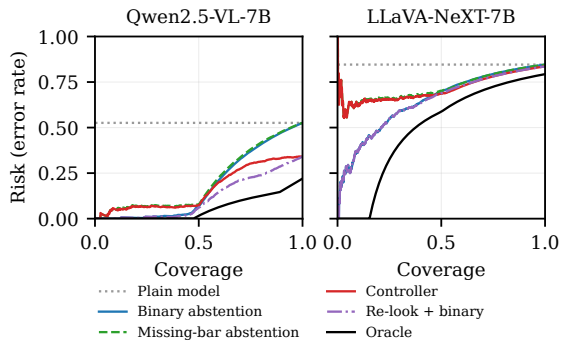


Figure 1: Risk versus coverage on synthetic value questions. The controller and Re-look + binary use the same repaired answers with different abstention rankings. At full coverage, the reduction in risk is the net gain from re-looking. Curves average over tied confidence scores.

because it flags misreads as well as missing-bar errors. Re-look + binary has the lowest AURC of the evaluated probe-based policies in both models (0.114 and 0.638).

Qwen selects verify in every fold. Of 222 fixes, 212 are Z answers converted to refusals, seven are fabrications, and only three are misreads; one correct refusal becomes wrong. Most gains are refusals on missing-bar questions, although the re-look was intended to correct misreads. The misread probe also responds to missing bars and therefore triggers this repair. LLaVA selects crop in every fold. It fixes 12 misreads and re-looks at 437 missing-bar questions without recovering any. With access to true classes, the oracle achieves lower AURCs of 0.056 and 0.497, so better policies are possible within this evaluation.

9 Discussion

E1 predicts wrong answers (AUROC 0.877 for Qwen, 0.798 for LLaVA; surface baselines 0.715 and 0.608). Real-chart probes distinguish misreads from computation errors (0.897 and 0.961), including after surface residualisation (0.918 and 0.920). This partly supports type separability for a different pair from the proposed misread/fabrication contrast. Question wording may still explain part of the separation: the surface controls do not remove question type.

The synthetic results depend on the model. Qwen's misread probe responds more strongly to misreads (0.887) than fabrications (0.594); LLaVA's responds more to fabrications (0.879) than misreads (0.776). Post-hoc gridline features with chart metadata exceed both probes on misreads (0.920 and 0.860). Synthetic misread probes also transfer weakly to ChartQA (0.597 and 0.557). We therefore cannot infer a general mechanism shared by the models from separability alone.

In pilot 3, the missing-versus-shown probes trained on absent category names detect out-of-range years (AUROC 1.000 and 0.911). Qwen's probe barely separates fabrication from correct refusal (0.521): its score tracks whether the chart supports answering, with little separation between these responses. Qwen refuses 102 of 300 out-of-range questions, compared with 15 of 600 absent-name questions. Its response differs between kinds of unsupported questions; chart design also changes between these pilots.

Repair effects depend on failure type in E4, yet the type-aware controller in E5 improves on binary abstention only for Qwen, mainly by prompting refusals. For LLaVA's missing-bar errors, policy abstention is the only effective option we tested. These results favour using answerability together with a general error score, and assessing the repair and abstention ranking as parts of the same policy.

10 Conclusion

To write (Yash): A short summary of what the experiments show.

Limitations, Challenges and Future Work

To write (Sanjith): Merge the labelling and synthetic-arm limitations with the E4/E5 text below.

E4 and E5 use synthetic bar charts and a single value-question template. Our crop requires bar geometry that the ChartQA setup does not provide. The balanced shown-bar and missing-bar design differs from the frequency of unsupported questions in natural chart use, so the controller AURCs should not be extrapolated to ChartQA. Rare failure classes need larger evaluation sets. Cross-fitting by figure still reuses the same synthetic pilots; we need an independent domain to test generalisation. We compare answer quality and coverage without optimising latency or inference cost. Future work should test repair selection on naturalistic unsupported questions and extend the analysis to node-link diagrams.

References

Shir Ashury-Tahan, Yifan Mai, Elron Bandel, Michal Shmueli-Scheuer, and Leshem Choshen. 2026. Er-

rorMap and ErrorAtlas: Charting the failure landscape of large language models. *arXiv preprint arXiv:2601.15812*.

Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, and 8 others. 2025. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*.

Anirudh Iyengar Kaniyar Narayana Iyengar, Tampu Ravi Kumar, Gaurav Najpande, Manan Suri, Dinesh Manocha, Puneet Mathur, and Vivek Gupta. 2026. DRAGON: A benchmark for evidence-grounded visual reasoning over diagrams. *arXiv preprint arXiv:2604.25231*.

Sai Akhil Kogilathota, Sripadha Vallabha E G, Luzhe Sun, and Jiawei Zhou. 2026. HALP: Detecting hallucinations in vision-language models without generating a single token. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 6067–6085.

Jinsong Li, Xiaoyi Dong, Yuhang Zang, Yuhang Cao, Jiaqi Wang, and Dahua Lin. 2026. Visual self-refine: A pixel-guided paradigm for accurate chart parsing. In *International Conference on Learning Representations (ICLR)*.

Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. 2024. LLaVA-NeXT: Improved reasoning, OCR, and world knowledge. https://llava-vl.github.io/blog/2024-01-30-llava-next/.

Jiaxin Liu, Ding Zhong, Yue Wang, Zhidong Yang, Zhaolu Kang, Guangyuan Dong, Qishi Zhan, Pengcheng Fang, and Aofan Liu. 2026. Dual-pathway circuits of object hallucination in vision-language models. *arXiv preprint arXiv:2605.13156*.

Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. 2022. ChartQA: A benchmark for question answering about charts with visual and logical reasoning. In *Findings of the Association for Computational Linguistics: ACL 2022*.

Anku Rani, Aparna Garimella, Apoorv Saxena, Balaji Vasan Srinivasan, and Paul Pu Liang. 2025. RADAR: A reasoning-guided attribution framework for explainable visual data analysis. *arXiv preprint arXiv:2508.16850*.

Subramanyam Sahoo, Vinija Jain, Aman Chadha, and Divya Chaudhary. 2026. Linear probes detect task format, not reasoning mode in language model hidden states. *arXiv preprint arXiv:2606.02907*.

Aojie Yuan, Zhiyuan Julian Su, Haiyue Zhang, Yi Nian, and Yue Zhao. 2026. Hidden error awareness in chain-of-thought reasoning: The signal is diagnostic, not causal. *arXiv preprint arXiv:2605.09502*.

Feiran Zhang, Yixin Wu, Zhenghua Wang, Xiaohua Wang, Changze Lv, Xuanjing Huang, and Xiaoqing Zheng. 2026. VIB-probe: Detecting and mitigating hallucinations in vision-language models via variational information bottleneck. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 23509–23521.

A Reproducibility

To write (Shrish): Hardware, library versions, data hashes, and the command for each result (README.md has the table).